ECG Dataset Compatibility Assessment

Metadata findings for acute STEMI and NSTEMI research

Research report for academic review and AI development planning
Analysis date 10 October 2026

Executive summary

The assessment identifies preparation requirements and evidence gaps across eleven ECG datasets. It supports selection and preprocessing discussions, but the present evidence does not confirm any dataset pair as ready for pooled acute STEMI, NSTEMI and non-MI supervised training. Clinical label validity remains a separate unresolved requirement.

The completed Colab run compared 55 unordered dataset pairs using published acquisition and structural descriptors. It generated 3 Low, 42 Moderate and 10 High minimum-effort ratings. All 55 ratings remained unconfirmed because important attributes were unresolved. The three levels correspond to the requested soft, medium and hard categories, with the question mark retained to show uncertainty. [1, 2]

Minimum effort	Pairs	Share of 55 pairs	Interpretation
Low or soft	3	5.5%	Few differences established in known fields
Moderate or medium	42	76.4%	Preparation or additional technical review indicated
High or hard	10	18.2%	Larger recording-context differences indicated

Table 1. Study-defined minimum-effort ratings. Percentages are rounded and sum to 100.1%. Every rating is unconfirmed; a Low rating does not establish common equipment or training suitability. Source: pairwise_assessment.csv in [2].

No ECG sample arrays or local dataset files were inspected in the default analysis. The Colab export records that optional metadata, header and HDF5 inspections were skipped. The findings therefore describe the curated documentation, rather than the contents or completeness of the downloaded PTB-XL and MIMIC folders. [1, 2]

The recommended next step is a bounded metadata and header validation of candidate sources, followed by clinical-label review and an agreed patient-level split plan. The AI developer can use the current tables to estimate preparation work and identify questions that must be resolved before training.

Study objective and analytical methods

Research question and target input

The question was which of the selected ECG datasets have compatible acquisition and representation attributes, and what work would be needed before combining them. The configured target was a diagnostic ECG with the standard twelve leads, 10 seconds of duration and a distributed sampling rate of 500 Hz. These are study settings, not a claim that this input is optimal for acute MI classification. [2: target_specification.json]

The wider project intends to compare a hierarchical MI versus non-MI model followed by STEMI versus NSTEMI classification with a flat three-class model. This report evaluates data preparation feasibility only. No model was trained and no accuracy, discrimination or clinical outcome was evaluated.

Evidence extraction and rating rules

The registry contains 38 attributes per dataset, covering sampling, leads, duration, equipment, calibration, filtering, containers, provenance, labels, linkage and access notes. Each field retains its value, evidence type and source pointer. Unknown means not established in this curated evidence set; it does not mean that the property is absent or that further documentation cannot be found. [2: attribute_dictionary.csv; attribute_evidence.csv]

For each pair, the engine applies explicit rules and reports the highest applicable effort level. Examples are shown in Table 2. Unknown fields do not trigger an equality match. Critical and additional unresolved attributes prevent confirmation even if the known minimum effort is Low. The scale is an operational study rule, not a clinically validated compatibility measure. [2: rulebook.csv; ecg_metadata_compatibility.py]

Level	Examples of applicable rules
Low	Container conversion, named-lead selection or reordering, and verified unit conversion
Moderate	Different sampling rates, duration or padding policies, device review, and calibration or preprocessing review
High	Different recording contexts, known filter or placement differences, or a required-lead mismatch
Unconfirmed	Missing evidence prevents confirmation of a Low, Moderate or High rating

Table 2. Summary of the rulebook. A rule describes a requirement; it does not execute preprocessing or validate the resulting signals.

Evidence coverage and exploratory clustering

Pairwise technical coverage is the fraction of 16 designated technical attributes known for both sources. It is a completeness measure, not a probability of successful pooling. The broader mixed-attribute distance uses available fields, normalized numeric differences, categorical mismatch and manufacturer-set Jaccard distance. Frequency and duration are log2-transformed; pairs below 60% comparable eligible weight are masked.

Clustering uses a separate complete set of four descriptors: distributed sampling rate, channel count, container and diagnostic/Holter context. Average linkage is applied to Gower-style dissimilarities. Sensitivity analyses remove the container feature or reduce its weight. No fixed number of clusters is imposed. [2: clustering_specification.json; mixed_distance_specification.json]

Dataset scope and documented characteristics

Table 3 identifies the selected releases. Counts are publisher-level registry entries and are not a measured inventory of available waveform files. Native and challenge editions must remain distinguishable. [2: dataset_profiles.csv; 3-12]

Dataset	Edition	ECGs	Hz	Leads	Duration s
PTB-XL	1.0.3 records500	21,799	500	12	10
MIMIC-IV-ECG	1.0	800,035	500	12	10
Ningbo	Challenge 1.0.3	34,905	500	12	10
Chapman-Shaoxing	Challenge 1.0.3	10,247	500	12	10
Georgia	Challenge 1.0.3	10,344	500	12	5-10
CPSC 2018	Challenge 1.0.3	6,877	500	12	6-144
CPSC-Extra	Challenge 1.0.3	3,453	500	12	6-144
SPH	Figshare v1	25,770	500	12	10-60
PTB Diagnostic	1.0.0 native	549	1000	15	Unknown
INCART	1.0.0 native	75	257	12	1800
CODE-15%	Zenodo 4916206	345,779	400	12	7-10

Table 3. Documented source characteristics. The 6-144 s CPSC range refers to the combined source description. CODE stored arrays include padding: 4096 samples at 400 Hz must not be interpreted as 10.24 s of observed ECG. PTB Diagnostic duration is unresolved in this registry.

Equipment and collection period

The registry attributes PTB-XL recordings to Schiller AG, MIMIC-IV-ECG to several manufacturers, SPH to a MedEx MECG-200, and PTB Diagnostic to a non-commercial PTB prototype. Manufacturer or model remains unknown for other profiles in the curated evidence. Brand membership alone does not prove equal acquisition settings, and a MIMIC cart identifier is not a verified manufacturer or model. [2: dataset_profiles.csv; 3, 4, 8, 9]

Collection periods are documented in the registry for PTB-XL (1989-1996), MIMIC-IV-ECG (2008-2019), SPH (2019-2020), and CODE-15% (2010-2016). Missing periods must not be inferred from publication years, challenge names or shifted patient timestamps. [2: attribute_evidence.csv]

Release and overlap precautions

The selected PTB-XL release has 21,799 ECGs; its 100 Hz and 500 Hz representations must not be counted as independent records. Native INCART contains 75 records, whereas the challenge subset has 74. Ningbo and Chapman-Shaoxing also appear together in a combined export. Related releases require an overlap review before datasets or splits are pooled. [2: DATASET_STRUCTURES.md; 3, 7, 10]

Pairwise preparation effort

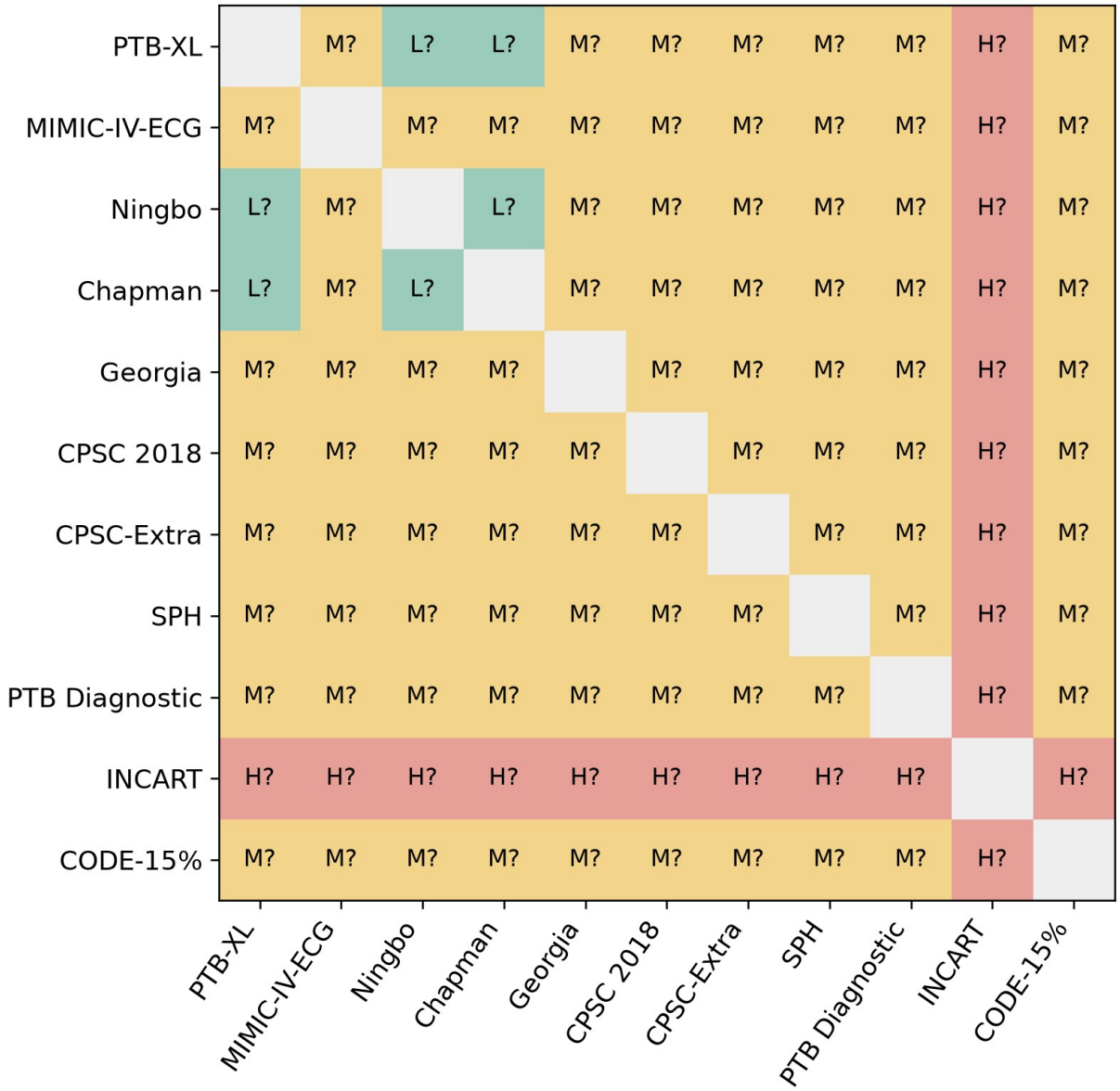


Figure 1. Pairwise minimum harmonization effort. L = Low or soft; M = Moderate or medium; H = High or hard. The question mark means the tier remains unconfirmed. The diagonal is not assessed. Chapman = Chapman-Shaoxing; SPH = Shandong Provincial Hospital. Redrawn directly from pairwise_assessment.csv in [2].

The three Low pairs are PTB-XL with Ningbo, PTB-XL with Chapman-Shaoxing, and Ningbo with Chapman-Shaoxing. Each has 43.75% coverage of the 16 technical comparison attributes. These ratings reflect few established structural differences; they do not establish the same device, filter settings, calibration or clinical labels.

All ten High pairs involve INCART, reflecting the diagnostic-versus-Holter recording-context rule. The remaining 42 pairs are Moderate. These are rule-based findings; no pooling experiment was performed. [2: pairwise_assessment.csv; pairwise_actions.csv]

Evidence completeness and unresolved attributes

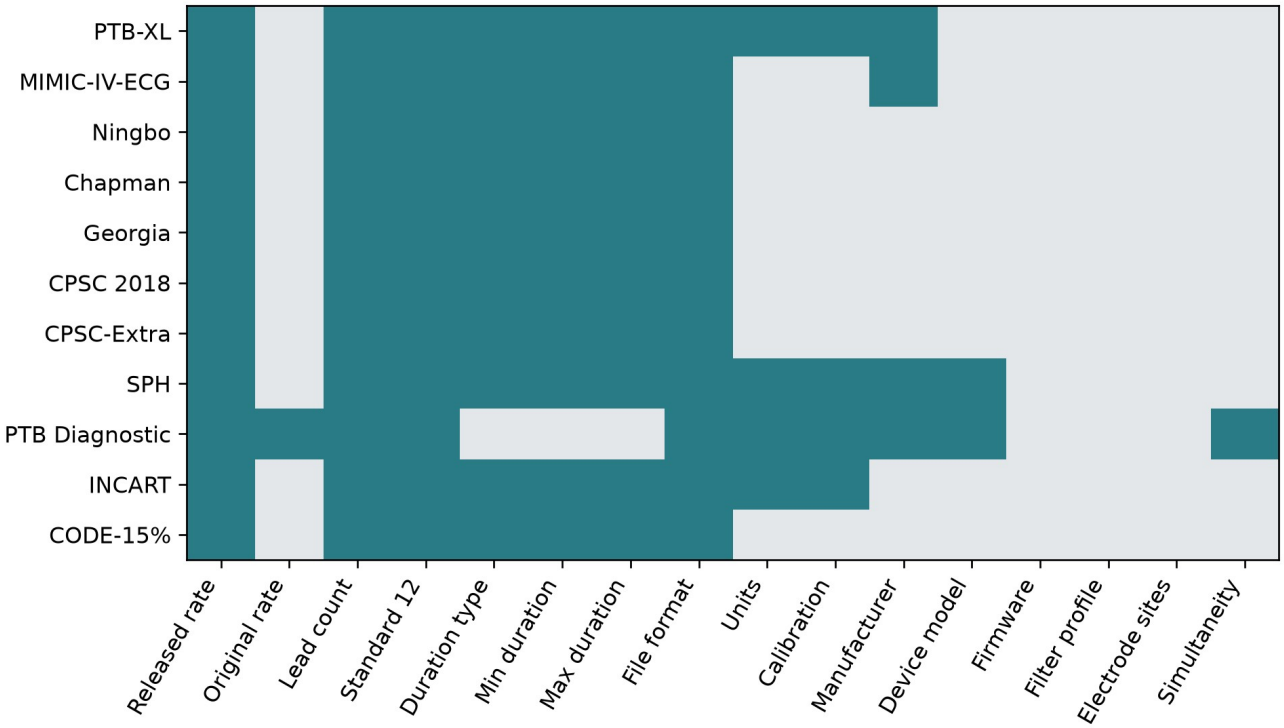


Figure 2. Technical evidence availability. Teal indicates a documented registry value; gray indicates an unknown in this evidence set. The figure does not imply record-level uniformity. Redrawn from registry_used.json in [2].

Evidence measure	Result	Interpretation
Documented entries across all 38 fields	195 / 418 (46.7%)	Includes technical and contextual fields
Unknown entries across all 38 fields	223 / 418 (53.3%)	Evidence gaps, not established absence
Mean pairwise technical coverage	42.5%	Across the separate 16-field subset
Median and range of pairwise coverage	43.75%; 25.0-62.5%	Descriptive completeness, not confidence

Table 4. Counts calculated from attribute_evidence.csv and pairwise_assessment.csv in [2]. Field-level and pairwise percentages have different denominators and must not be combined.

Device model, firmware, filter profile, electrode placement and simultaneous-acquisition information are sparsely established. Every pair has unresolved attributes, so the analysis cannot confirm equipment equivalence. The evidence table should guide targeted source review and optional file-header checks.

The treatment of unknowns is intentionally conservative, but the result also depends on the chosen confirmation requirements. Some missing attributes may be difficult to recover from public data. A supervisor-reviewed protocol should state how unresolved information affects inclusion, rather than interpreting every unknown as a permanent exclusion.

Preparation requirements for the AI developer

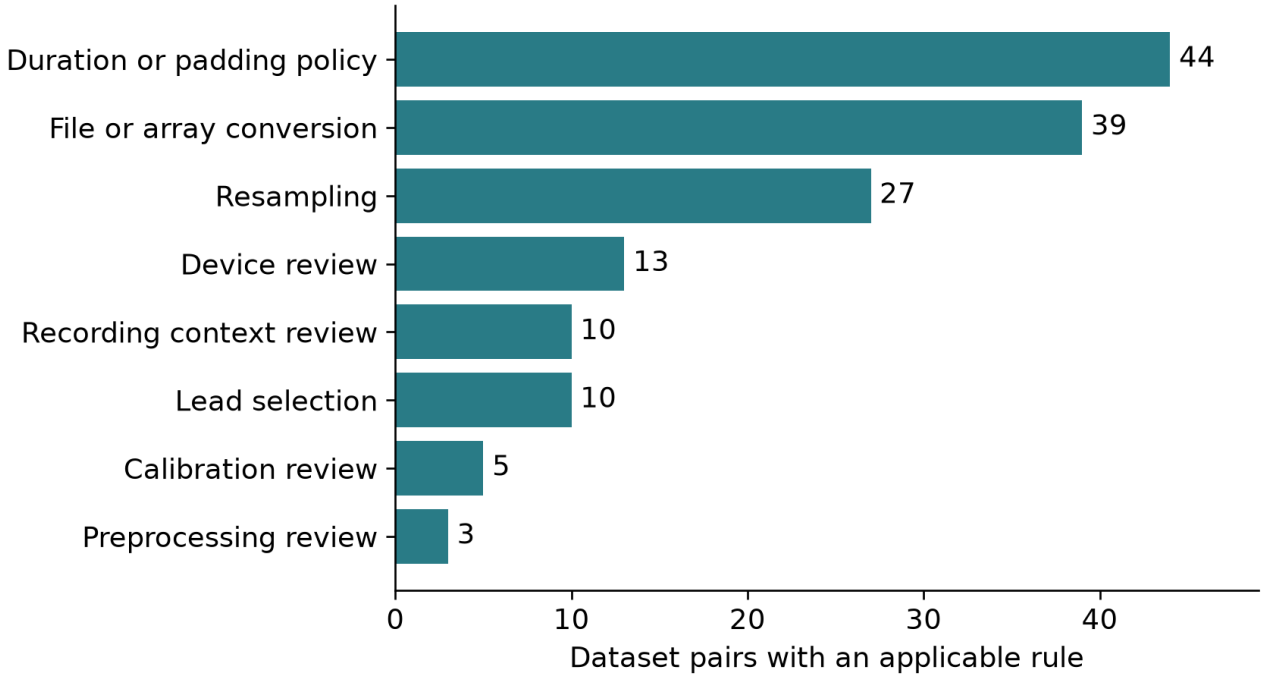


Figure 3. Number of pairs triggering each preparation rule. Requirements can overlap within a pair. A zero or absent rule count does not establish that the corresponding property is equal; it may be unknown. Redrawn from pairwise_actions.csv in [2].

Duration or padding policy is triggered for 44 pairs, container conversion for 39, and resampling for 27. Device review is indicated for 13 pairs. Lead selection and recording-context review each occur in 10 pairs; calibration and preprocessing review occur in 5 and 3 pairs, respectively. These counts prioritize investigation and specification work; they are not estimates of developer hours.

Worked example of PTB XL and MIMIC IV ECG

Both selected profiles document twelve leads, 10-second duration and a 500 Hz distributed rate. Their pair receives a Moderate minimum-effort rating with 50% technical evidence coverage. The known rules indicate device and preprocessing review. MIMIC physical units and calibration are unresolved in this registry, and several device, filtering and acquisition details remain unconfirmed. [2: pairwise_assessment.csv]

The appropriate developer action is to verify header calibration and lead order, inspect available device/filter metadata, and document the common input representation. A successful schema conversion would not settle clinical eligibility. The pair should remain provisional until its label evidence and patient/encounter linkage are reviewed.

Data preservation

A proposed preparation specification should retain source identifiers, release version, patient grouping, native and target rates, original duration and any padding mask. Record each transformation and its justification. Do not overwrite original files when implementing a common model input.

Exploratory clustering and sensitivity

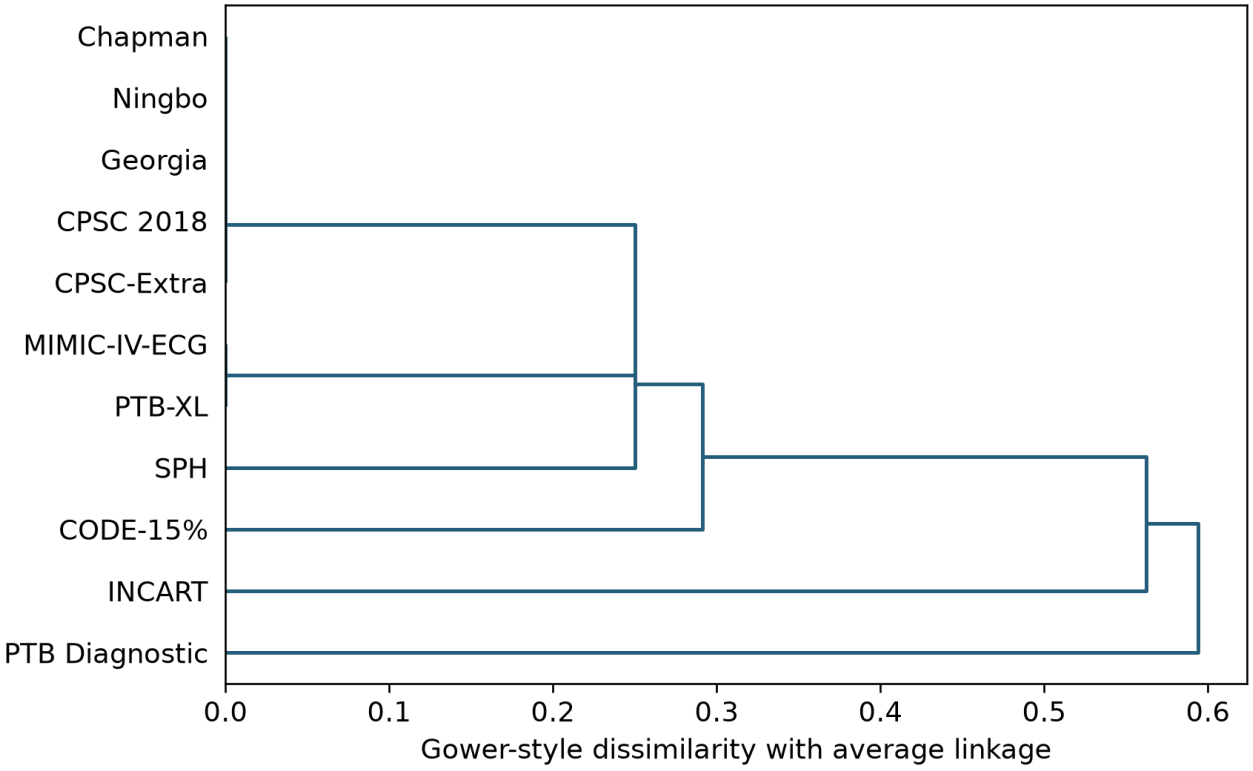


Figure 4. Primary clustering on distributed sampling rate, channel count, container and recording context. Some datasets have identical values on these four fields, producing zero-distance ties. Device, filter and clinical-label similarity cannot be inferred. Redrawn from linkage_core_equal.csv in [2].

Scenario	Distance correlation	Tree distance correlation	Mean distance change
Equal feature weights	1.000	1.000	0.000
Container omitted	0.937	0.949	0.157
Container downweighted	1.000	0.989	0.109

Table 5. Spearman correlations against the equal-weight baseline. Tree distance refers to cophenetic distance. Values are deterministic sensitivity summaries, not patient-level inference or statistical significance tests. Source: clustering_sensitivity.csv in [2].

Removing container gives distance and cophenetic correlations of 0.937 and 0.949. Downweighting it gives 1.000 and 0.989. This shows broad agreement within the chosen descriptors, while absolute distances change. It does not establish that the clusters generalize to unmeasured hardware, filters or populations.

The cluster structure is constrained by a small, coarse feature set. Several 500 Hz twelve-lead diagnostic sources become identical when container is omitted. Use these trees as a descriptive view alongside the evidence matrix and pairwise rules, rather than as independent confirmation of compatibility.

Dataset dispositions and immediate actions

The target-readiness rules identify three technical candidates for harmonization and eight sources to hold for evidence or domain review. Candidate status only means the selected technical blockers did not trigger; unresolved fields remain. All eleven sources retain unverified acute three-class labels. [2: target_readiness.csv]

Dataset	Current technical status	Immediate preparation or review
PTB-XL	Candidate	Check acquisition gaps; review acute timing and subtype evidence
MIMIC-IV-ECG	Hold	Verify amplitude calibration; review clinical linkage and adjudication
Ningbo	Hold	Verify calibration and acquisition details; review label meaning
Chapman-Shaoxing	Hold	Verify calibration; distinguish original and challenge exports
Georgia	Hold	Verify calibration and 5-10 s window policy; review patient linkage
CPSC 2018	Hold	Verify calibration; define policy for variable durations
CPSC-Extra	Hold	As above; review overlap with related CPSC exports
SPH	Candidate	Define duration policy; review AHA statements and acquisition gaps
PTB Diagnostic	Candidate	Select standard twelve leads; plan 1000-to-500 Hz conversion; verify duration
INCART	Hold	Review Holter domain and window-level eligibility before inclusion
CODE-15%	Hold	Verify amplitude scale, padding interpretation and target-label availability

Table 6. Technical dispositions from target_readiness.csv, with practical review actions derived from the reported gaps. Candidate does not mean approved for training. Source: [2].

Clinical labels require an independent decision

For each proposed target, retain the original label, its source, ECG timing, supporting clinical evidence and reviewer decision. Broad MI statements, machine text, ST/T-change labels and rhythm codes are not converted to acute STEMI/NSTEMI outcomes in this study. Absence of an MI code is not used to create non-MI. [2: METHODS_AND_RESULTS.md]

A dataset without verified target labels may still be considered for pretraining after technical and access review. That possible role must be recorded separately from supervised acute-MI training or external clinical validation. Final inclusion decisions belong in a reviewed study protocol.

Data structures and developer handoff

The structures below are examples for the selected editions. They were not inspected on the local filesystem during this run. Path existence, sample completeness and header-to-signal consistency remain to be checked before data loading. [2: DATASET_STRUCTURES.md]

Source group	Documented structure	Keys or interpretation
PTB-XL	ptbxl_database.csv; scp_statements.csv; records500/.../{record}.hea and .dat	ecg_id, patient_id and filename_hr; records100 is another representation
MIMIC-IV-ECG	record_list.csv; machine_measurements.csv; files/p.../p.../s.../{study}.hea and .dat	study_id identifies the ECG; subject_id groups patients; cart_id is not a model
Ningbo, Chapman, Georgia, CPSC sources	{record}.hea with {record}.mat; optional subfolders	Selected challenge edition; val array is leads x samples
SPH	metadata.csv; code.csv; per-record .h5 files	Patient_ID; documented arrays are leads x variable samples
PTB Diagnostic	patient001/s0010_re.hea and .dat	Patient folder for grouping; standard twelve plus Frank X/Y/Z
INCART	I01.hea; I01.dat; I01.atr	Header calibration; .atr stores beat annotations
CODE-15%	exams.csv; exams_part*.hdf5 with exam_id and tracings	Join on exam_id, not row position; tracings shape N x 4096 x 12

Table 7. Developer structure guide. Example record names are illustrative. Full example layouts and provenance notes are in DATASET_STRUCTURES.md in [2].

Files to send with this report

Provide the executed notebook and the results ZIP, including target_specification.json, dataset_profiles.csv, attribute_evidence.csv, pairwise_assessment.csv, pairwise_actions.csv, target_readiness.csv, rulebook.csv and run_provenance.json. These preserve details that a formatted report cannot display in full.

Agree a data contract before implementation: lead names and order, physical units and verified conversion, target frequency, window and padding policy, patient grouping, inclusion rules, reviewed labels, and source-version identifiers. Keep unresolved choices in an issue log with an evidence source, proposed owner and decision status.

Limitations and recommended next stage

Limits of the current findings

This was a documentary assessment with no local file inspection. Published attributes may describe a whole release, a combined source or an export convention rather than every record. The registry and its rulebook determine the result; no independent clinical scale or empirical model comparison validates the effort tiers. Missing attributes could alter both ratings and preparation requirements.

The evidence coverage measure gives equal presence credit to the designated fields, including attributes shared across many sources. It does not measure information content or confidence. Pairwise comparisons are not independent observations because each dataset appears in ten pairs. The clustering features are coarse and incomplete with respect to equipment, preprocessing, population and labels.

Waveform quality, sample-level calibration accuracy, corruption, signal duplicates, population shift and clinical generalization were not assessed. Patient counts, acute-event prevalence, download completeness and train/test independence cannot be concluded from these outputs. The package contains no trained-model evaluation. [1, 2]

Recommended work for the data analyst

1. Validate bounded local observations. Enable metadata and header checks for PTB-XL and available MIMIC data. Record selection rules, files attempted, readable headers, missing files and conflicts. Label prefix previews and convenience samples explicitly.
2. Resolve priority evidence gaps. Start with units, gain/baseline, lead order, device semantics and filter metadata. Retain source URLs, release, locator, reviewer and date. Do not generalize a single-header finding to an entire release.
3. Prepare a label-readiness table. Link original labels and ECG timing to supporting evidence for review with the clinical supervisor. Preserve uncertain and unresolved cases instead of forcing three-class assignments.
4. Propose leakage controls. Group repeated ECGs by patient where defensible, document encounter linkage, and review alternate resolutions, original/challenge exports and derivative datasets for overlap.
5. Freeze a shared comparison protocol. Once labels and inclusion are reviewed, use the same eligible cohort, patient splits and input specification when comparing hierarchical and flat models. Plan source-specific results and external validation with the AI team.

Decision for the next meeting

Use the report to approve a targeted evidence-resolution and metadata-validation phase. The current results justify preparation planning and a shortlist for further review; they do not yet justify an unconditional pooled training cohort.

Reproducibility record and references

Run item	Recorded value
Run timestamp	10 October 2026 at 09:11:04 UTC
Engine and Python	Engine 1.0.0; Python 3.13.16
Core packages	NumPy 2.1.3; pandas 2.2.3; SciPy 1.16.3; Matplotlib 3.10.0
Main analysis duration	10.256 seconds; excludes any later review or report preparation
Inspection scope	0 waveform arrays read; 0 local dataset files inspected by default

Table 8. Values from run_provenance.json in [2]. The saved registry digest was recomputed during report preparation and matched the recorded value, confirming internal registry identity rather than external factual validity.

Analysis artifacts

[1] MI_Metadata_Compatibility_Colab.ipynb - Colab.pdf. Executed notebook export, 34 pages, printed 10 October 2026. Results and scope: pp. 11, 24-25; optional checks skipped: pp. 28, 30-31.

[2] metadata_compatibility_v1.zip. Colab results archive dated 10 October 2026. Tables, registry, source register, engine and provenance used throughout this report. All report figures were redrawn from archived data or linkage matrices; ratings and clustering were not redefined.

Dataset references recorded in the registry

[3] PTB-XL version 1.0.3. https://physionet.org/content/ptb-xl/1.0.3/

[4] MIMIC-IV-ECG version 1.0. https://physionet.org/content/mimic-iv-ecg/1.0/

[5] PhysioNet Challenge 2021 source descriptions. https://moody-challenge.physionet.org/2021/

[6] Challenge 2021 distribution version 1.0.3. https://physionet.org/content/challenge-2021/1.0.3/

[7] Ningbo and Chapman-Shaoxing source history. https://physionet.org/content/ecg-arrhythmia/1.0.0/

[8] Liu et al. SPH data descriptor, Scientific Data, 2022. https://doi.org/10.1038/s41597-022-01403-5

[9] PTB Diagnostic version 1.0.0. https://physionet.org/content/ptbdb/1.0.0/

[10] INCART version 1.0.0. https://physionet.org/content/incartdb/1.0.0/

[11] CODE-15% release 4916206. https://zenodo.org/records/4916206

[12] Author ECG preprocessing documentation. https://github.com/antonior92/ecg-preprocessing/blob/main/README.md